

A.1 Kinematics

Past Paper Answers

These are past paper answers selected from past papers (2016 - 2022) that relate to the topic A.1 Kinematics.

The associated questions are in the separate accompanying document

Answer 1

A

Answer 2

C

Answer 3

C

Answer 4

determines component correctly / $15 \sin 50$ seen ✓ $s = \frac{(15 \sin 50)^2}{2 \times 9.81} = 6.7 \text{ m}$ ✓ correct reasoning consistent with candidate data ✓	Allow method via $v = u + at$. Allow use of $g = 10 \text{ m s}^{-2}$, gives 6.6 m and 8.1 m. Allow [2 max] for use of $15 \cos 50$, gives 4.7 m and 6.2 m. Allow [1 max] (as MP2) if 13 m is obtained due to use of 15 m s^{-1} rather than $15 \sin$ or $15 \cos 50$. If no unit given, assume metre.	3
--	--	----------

Answer 5

C

Answer 6

C

Answer 7

D

Answer 8

B

Answer 9

B

Answer 10

correct use of kinematic equation/equations ✓ 148.5 or 149 or 150 «m» ✓	<i>Substitution(s) must be correct.</i>	2
--	---	----------

Answer 11

D

Answer 12

C

Answer 13

A

Answer 14

D

Answer 15

use of conservation of energy OR $v^2 = u^2 + 2as$ ✓ $v = \sqrt{2 \times 60.0 \times 9.81} = 34.3 \text{ «ms}^{-1}\text{»}$ ✓		2
---	--	----------

Answer 16

C

Answer 17

A

Answer 18

A

Answer 19

A

Answer 20

C

Answer 21

B

Answer 22

B

Answer 23

C

Answer 24

D

Answer 25

A

Answer 26

B

Answer 27

D

Answer 28

C

Answer 29

D

Answer 30

C